A FINITE TORSION FREE MODULE OVER A DOMAIN IS A SUBMODULE OF A FINITE FREE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of the cards already published from this chapter, so that the batch selector treats them as one source instead of deferring them as a same-source clash.

Allowed standard prerequisites (editorial).

- A submodule of a free module over a domain is torsion free. used at proof line 4871 for the easy implication. Gives the implication from a submodule of R^{n} to torsion freeness.
- Torsion freeness is equivalent to injectivity of M to M tensor K. the discussion printed at source lines 4794-4799 of the same section. Identifies the image of M inside M tensor K, which is the space in which the free submodule is found.
- M tensor K is a finite dimensional vector space for a finite module M over a domain. used at proof lines 4872-4875 to choose a basis e_1 to e_r and generators x_1 to x_n. Produces the finite basis and the expressions x_i = sum (a_ij/b_ij) e_j that the proof clears denominators in.

Source reference. Original lemma 0AUU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13662–13662. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a domain. Let M be a finite R-module. Then M is torsion free if and only if M is a submodule of a finite free module.*

Proof. If M is a submodule of $R^{\oplus n}$, then M is torsion free. For the converse, assume M is torsion free. Let K be the fraction field of R. Then $M \otimes_R K$ is a

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

finite dimensional K-vector space. Choose a basis $e_1, \ldots, e_r$ for this vector space. Let $x_1, \ldots, x_n$ be generators of M. Write $x_i = \sum (a_{ij}/b_{ij}) e_j$ for some $a_{ij}, b_{ij} \in R$ with $b_{ij} \neq 0$. Set $b = \prod_{i,j} b_{ij}$. Since M is torsion free the map $M \to M \otimes_R K$ is injective and the image is contained in $R^{\oplus r} = Re_1/b \oplus \ldots \oplus Re_r/b$. $\square$